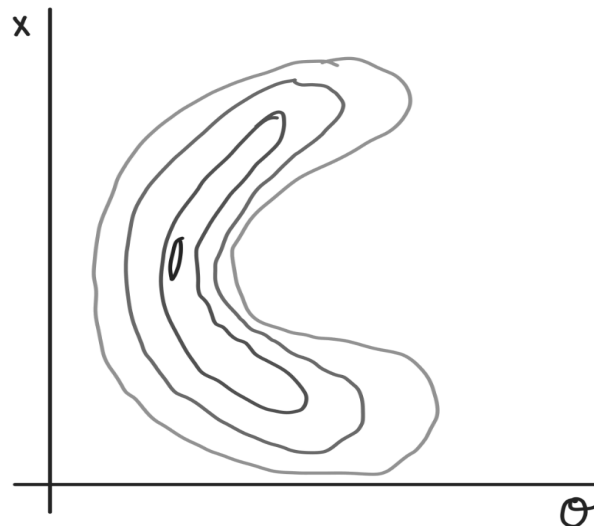


## Protocol Data Literacy Midterm Exam 13.12.2024

### Exercise 1



1. In the figure, sketch  $P(x)$  -> Marginalize over  $\theta$
2. In the figure, sketch  $P(\theta)$  -> Marginalize over  $x$
3. Sketch the mean  $\mu(x) := E(\theta|x)$
4. For  $\theta_1 = 0.5$  and  $\theta_2 = 3$ , sketch  $P(x|\theta_i)$  inside the figure.
5. Are  $\theta$  and  $X$  independent?

### Exercise 2

You are accused of using AI tools for your thesis. The professor used a tool to detect this, which has an 80% chance of a positive result given that AI was being used, which is why he accuses you.

Luckily, you are familiar with statistics, and know that 20% of students use AI assistance for their thesis and the chance of the tool detecting AI given that it was not used, is also 20%. Calculate the probability of using AI given a positive test result.

Multiple choice questions, similar to this:

1. Which of the following statements about the conditional distribution  $P(X|Y)$  are correct? (assume that  $X$  and  $Y$  are not independent)

☐  $p(X|Y) = \frac{p(X,Y)}{\int p(X,Y)dY}$

☐  $p(X|Y) = \frac{p(X,Y)}{\int p(X,Y)dX}$

☐  $p(X|Y) = \frac{p(X)}{p(Y)}$

☐  $\int p(X|Y)dY = 1$

☐  $\mathbb{E}_{p(Y)}[p(X|Y)] = p(X)$

☐  $\mathbb{E}_{p(X)}[p(X|Y)] = p(X)$

+ another MC question for the case where  $X$  and  $Y$  are independent

This question but with  $b^2$  instead of  $b$

3. Assume that we have a  $n$ -dimensional Gaussian random variable  $U$  with mean  $\mathbb{E}[U] = \mu$ . We define a second random variable  $X$  such that  $X = AU + b$ , where  $A \in \mathbb{R}^{n \times n}$ ,  $b \in \mathbb{R}^n$ . What is the expectation  $\mathbb{E}[X]$ ?

☐  $\mu$

☐  $A\mu$

☐  $A\mu + b$

☐  $A\mu A^T$

### Exercise 3

- 1) Show that the variance of a Bernoulli distribution is  $p^*(1-p)$ , given  $\mathbb{E}[x] = p$
- 2) Then the following exercise was very close to this one, just different numbers so you did not need a calculator

### 1. EXAMPLE Question — The importance of Importance Sampling

Let's say we want to understand the relationship between attending lectures and passing the respective exams throughout the University of Tübingen. We assume that attending and passing are binary events (a student either attends all the lectures of a course or skips them altogether, and either passes—with sufficient—or fails the exam—not sufficient—, independent of the other students). In order to understand this relationship, we ran a survey in the Data Literacy course after grading the exam by asking the students whether they attended the lectures. 100 students responded in the following way:

Table 1: Survey results.

|                             | Attend ( $A$ ) | Skip ( $\neg A$ ) |
|-----------------------------|----------------|-------------------|
| Sufficient ( $S$ )          | 56             | 5                 |
| Not sufficient ( $\neg S$ ) | 10             | 29                |

- (a) Using the Sampling Estimator and Table 1, estimate the probabilities  $P(S|\neg A)$  and  $P(\neg S|A)$ .

What is the variance of the estimators? Note that, as we have seen in the lecture, in order to estimate the variance of the estimators, we need to make the rather unrealistic assumption that we know the expected  $P(S|\neg A)$  and  $P(\neg S|A)$ , so we will assume here that we know that  $P(S|\neg A) = 0.1$  and  $P(\neg S|A) = 0.2$ .

*Solution:* Defining  $X$  as a Bernoulli random variable, we use the sampling estimator to estimate  $P(S|\neg A)$  and  $P(\neg S|A)$  as

$$\hat{P}(S|\neg A) = \frac{1}{|\neg A|} \sum_{e \in \neg A} X_{e \in S} = \frac{1}{34} 5 = \frac{5}{34} \approx 0.147$$

and

$$\hat{P}(\neg S|A) = \frac{1}{|A|} \sum_{e \in A} X_{e \in \neg S} = \frac{1}{66} 10 = \frac{10}{66} = 0.15.$$

From the description we know  $P(S|\neg A) = 0.1$  and  $P(\neg S|A) = 0.2$  and therefore can compute variances as

$$\begin{aligned} \text{var}(P(S|\neg A)) &= \text{var}\left(\frac{1}{|\neg A|} \sum_{e \in \neg A} X_{e \in S}\right) = \frac{1}{|\neg A|^2} \sum_{e \in \neg A} \text{var}(X_{e \in S}) \\ &= \frac{|\neg A|}{|\neg A|^2} P(S|\neg A)(1 - P(S|\neg A)) = \frac{34}{34^2} \cdot 0.1 \cdot 0.9 \approx 0.0027 \end{aligned}$$

- (b) Overall, at the University of Tübingen, 50% of students attend lectures. Using the data in Table 1 and importance sampling, estimate the probability that students throughout the university pass a course  $P(S)$ .

*Solution:* After estimating  $Q(A) = \frac{66}{100}$  and  $Q(\neg A) = \frac{34}{100}$  from Table 1, we can calculate

$$\begin{aligned} \widetilde{P(S)} &= \widetilde{P(S \cap A)} + \widetilde{P(S \cap \neg A)} \\ &= Q(S \cap A) \cdot \frac{P(A)}{Q(A)} + Q(S \cap \neg A) \cdot \frac{P(\neg A)}{Q(\neg A)} \\ &= \frac{56}{100} \cdot \frac{0.5}{0.66} + \frac{5}{100} \cdot \frac{0.5}{0.34} \approx 0.498 \end{aligned}$$

5) Name two assumptions you have to make for your estimation to work, that might not be true

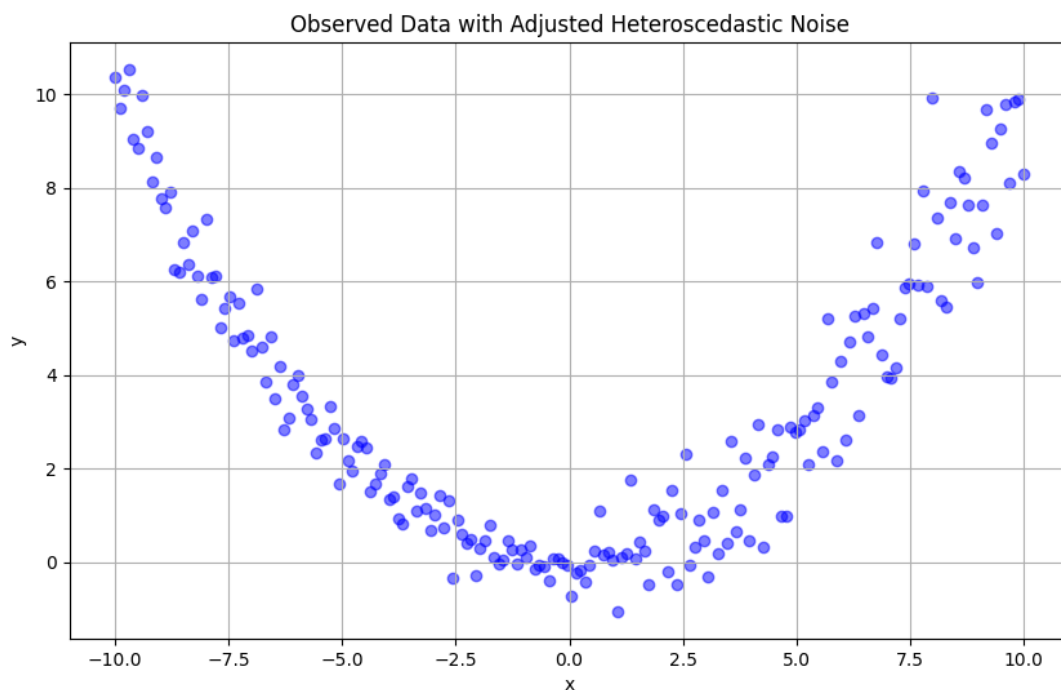
### Exercise 4

The reception of the calls in the call center can be described with the exponential distribution, where parameter lambda is a reception rate. For the data  $D = \{x_1, \dots, x_n\}$  calculate MLE estimator for the exponential distribution with

$$p_{\text{exp}}(x) = \lambda e^{-\lambda x}$$

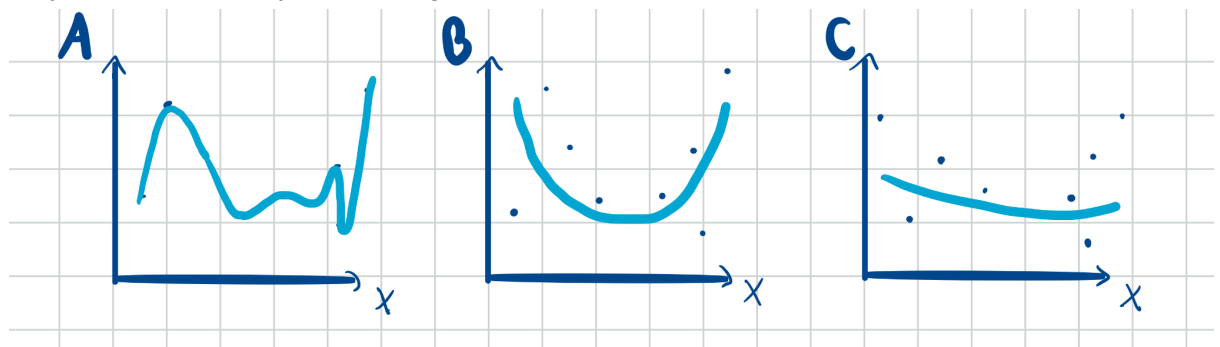
- 1) Write down the likelihood function
- 2) Write down the derivative of the log likelihood function
- 3) Find MLE estimator
- 4) Having the  $D = \{15, 5, 10\}$  minutes calculate the expected number of calls per hour based on the MLE estimator

### Exercise 5



(It was more extreme in the exam so you could see it even clearer)

1. Choose a set of three basis functions that fit this example
2. Why can we not apply Linear Regression here? which assumptions are violated?



What is the regularisation parameter for the plots ? (assign them)

0.01

10000

10

What is the mean squared error? (0.76, 5, 19)

How would you choose the regularisation parameter in practice?

### Exercise 6

Hypothesis testing: Do people who eat coriander live longer? We conduct an experiment and sample from a group of coriander eaters and a group of non-coriander eaters. From this sample, we get two means  $\mu_1$  (coriander eaters) and  $\mu_2$ .

- a) What would be a suitable  $H_0$  in this case?
- b) We perform a test which yields  $p=0.01$ . Decide whether the following statements are true or false. For the false statements explain why they are incorrect.
  - i) We have disproved the null hypothesis.
  - ii) We reject the null hypothesis.
  - iii) If we would repeat the experiment 1000 times then in 990 cases we would get significant results.
  - iv) Our result shows that there is a difference between the groups.
  - v) Our data provides strong evidence against the null hypothesis.
  - vi) There is strong evidence that eating coriander makes you live longer